STATISTICAL QUALITY CONTROL | DMAIC CASE STUDY

# Statistical Process Control for Personal Habits

*Measuring, diagnosing, and monitoring ten daily habit defects over 48 days with Pareto analysis, root-cause analysis, and control charts*

**Raja Darshini Rajamani** | STAT 513 Final Project | Data collected February 20 to April 8, 2025

| | |
|---|---|
| **Objective** | Quantify a personal daily routine as a quality process, find the defects that matter most and what drives them, and test whether the routine is statistically stable. |
| **Methods** | DMAIC framework, check sheet, Pareto analysis, Ishikawa (fishbone) diagram, stratification, phi correlation, P and NP charts, $\bar{X}$ and S charts, I-MR chart, CUSUM, EWMA, moving averages |
| **Tools** | Microsoft Excel (data capture), Python with pandas, SciPy, and matplotlib (analysis and charts) |

| 48 | 480 | 44.8% | 0 |
|---|---|---|---|
| days observed | habit checks | baseline defect rate | out-of-control signals |

## Executive summary

For 48 consecutive days I recorded ten daily habits as pass/fail checks and treated each missed habit as a process defect. Using the DMAIC framework, I asked three questions: which defects matter most, what drives them, and is the routine statistically stable?

**Key findings**

1. **Defects are concentrated.** Three habits (neglected personal projects, excessive social media, and not applying for jobs) produce 50% of all 215 defects, and six habits produce 80%.
2. **The routine is stable but not yet capable.** Every control chart (P, NP, $\bar{X}$ and S, I-MR, CUSUM, EWMA) shows the process in statistical control around a 44.8% defect rate (about 447,900 DPMO, or 1.6 sigma). Variation is common-cause, so lasting improvement requires changing the system rather than reacting to individual bad days.
3. **The weekly schedule is the strongest driver.** Weekdays average 4.9 defects versus 3.4 on weekends (Mann-Whitney $p = 0.004$). Skipped exercise (65% vs. 7% of days) and short sleep (62% vs. 7%) are almost entirely weekday problems.
4. **Habits move in clusters.** Short sleep co-occurs with skipped exercise ($r = 0.46$) and lateness ($r = 0.37$). Excessive social media is *negatively* related to short sleep ($r = -0.42$): both track how much free time a day contains, which contradicted my initial assumption that scrolling was costing me sleep.
5. **Breaks shift the mix, not the system.** Spring Break produced the lowest-defect stretch (3.6 per day) and the return to classes the worst single day (8 of 10 on March 26). The CUSUM registered a $-3.0\sigma$ excursion that stayed within its decision interval and reset within a week.

**Recommendation.** Prioritize two system changes: a protected weekday career block (targets two of the top three Pareto defects) and a sleep-first weekday schedule (targets the sleep, exercise, and punctuality cluster), monitored with a P chart and EWMA against this baseline.

# 1. Define

## 1.1 Problem statement

A daily routine behaves like a production process: each day produces an output, a set of habits that were either completed or missed. Without measurement, it is difficult to tell whether a bad week reflects normal variation or a real change, and improvement efforts tend to target whatever feels most urgent rather than what matters most. This project applies quality engineering methods to personal habit data so that improvement can be targeted with evidence.

## 1.2 Project charter

**Table 1.** Project charter

| Element | Definition |
|---|---|
| **Goal** | Establish a baseline defect rate, identify the largest contributors and their drivers, and assess process stability. |
| **Scope** | Ten habits, recorded daily from February 20 to April 8, 2025 (48 days). |
| **Unit and opportunities** | One day = one subgroup of 10 habit checks; 480 opportunities in total. |
| **Defect** | A habit check that fails its operational definition (Table 2). |
| **Primary metric** | Daily defect proportion, p = defects ÷ 10. |
| **DMAIC status** | Define, Measure, and Analyze completed with data. Improve and Control are proposed as the next cycle (Sections 5 and 6). |
| **Out of scope** | Severity weighting, mood and stress levels, and exam dates (listed as future work). |

## 1.3 Critical-to-quality habits and operational definitions

Each habit was given a measurable threshold so that recording required no judgment calls. Habits were grouped into five themes to support later analysis.

**Table 2.** Operational definitions of the ten habit defects

| # | Defect | Operational definition (recorded as a defect when...) | Theme |
|---|---|---|---|
| 1 | Late to class/work | Arrived after the scheduled start time | Time management |
| 2 | Excessive social media use | Spent more than 1 hour on TikTok and Instagram | Time management |
| 3 | Skipped exercise | No physical activity (dance, swimming, weightlifting, badminton, or yoga) | Health |
| 4 | Did not get enough sleep | Slept less than 7 hours | Health |
| 5 | No social interaction | Did not meet or spend time with friends | Relationships |
| 6 | Skipped meals | Ate fewer than 2 balanced meals | Health |
| 7 | Did not check in with parents | Did not call or update my parents about my day | Relationships |
| 8 | Did not apply for jobs | Applied for fewer than 5 jobs | Career |
| 9 | Neglected personal projects | Spent less than 30 minutes on personal projects | Career |
| 10 | Did not read a book | Read for less than 20 minutes (morning or night) | Personal growth |

# 2. Measure

## 2.1 Data collection plan

- **Instrument.** An Excel check sheet with one row per habit and one column per day; each cell is TRUE (defect) or FALSE (no defect).
- **Timing.** Recorded once per day at midnight for the day just ended, so every observation covers the same window.
- **Consistency.** Numeric thresholds (Table 2) remove ambiguity, and daily totals were cross-checked between the check sheet and the analysis sheet.
- **Completeness.** All 48 days and 480 checks were recorded; there are no missing values.

## 2.2 Baseline performance

**Table 3.** Baseline process performance

| Measure | Value | Notes |
|---|---|---|
| Defects / opportunities | 215 / 480 | 10 checks per day for 48 days |
| Defects per opportunity ($\bar{p}$) | 0.448 | Baseline defect rate of 44.8% |
| Defects per million opportunities | 447,917 | Process sigma of about 1.6 (1.5σ shift convention) |
| Mean defects per day | 4.48 (SD 1.58) | Median 4; no defect-free days |
| Best days | 1 defect | March 9 and March 23 (both Sundays) |
| Worst day | 8 defects | March 26, the Wednesday after Spring Break |

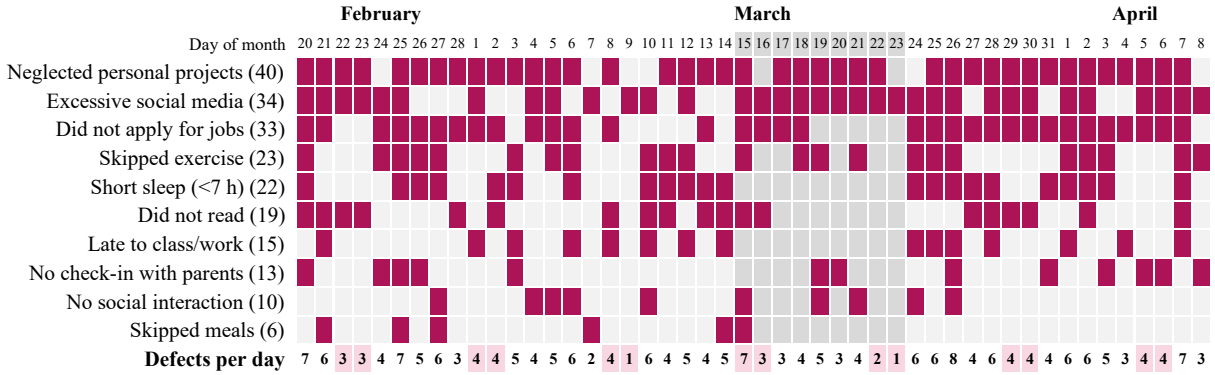



**Figure 1.** Defect map. Each cell is one habit on one day (dark pink = defect), sorted by frequency with totals in parentheses. Gray columns mark Spring Break (March 15 to 23), and weekend totals in the bottom row are shaded light pink.

The defect map makes two patterns visible before any statistics are run. First, three habits are missed on most days and form near-solid rows. Second, the routine changes character during Spring Break: short sleep and lateness disappear entirely, while excessive social media becomes a daily occurrence.

# 3. Analyze

## 3.1 Pareto analysis

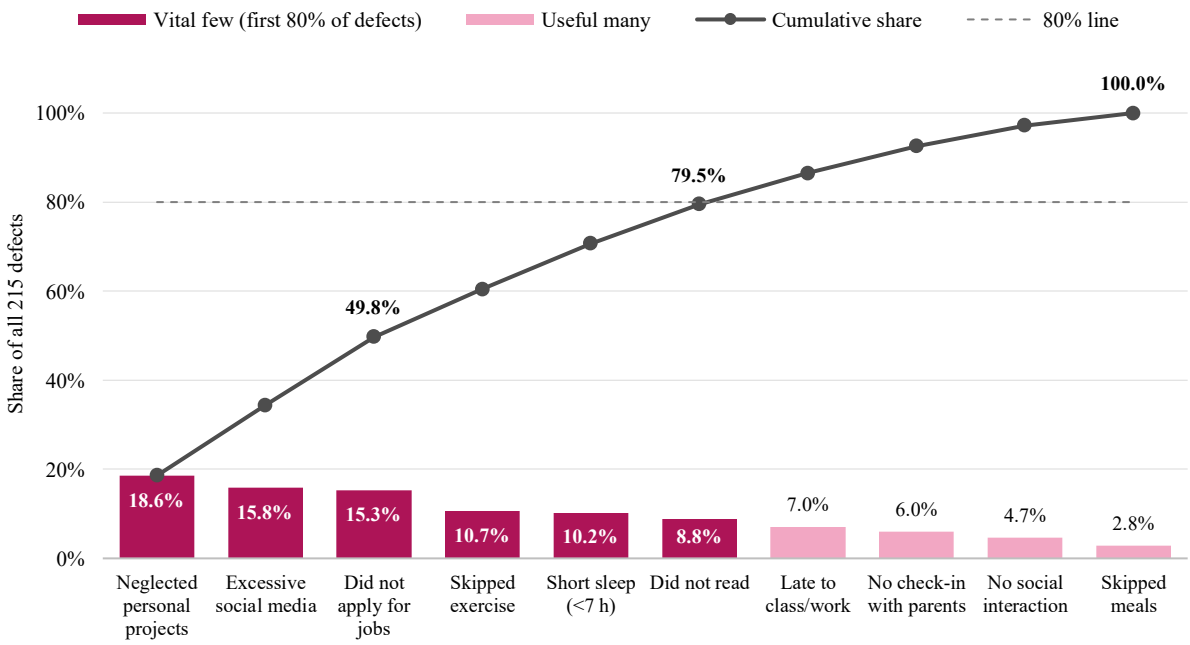



**Figure 2.** Pareto chart of all 215 defects. Bars show each habit's share of the total; the line shows the cumulative share.

The distribution follows the Pareto principle closely. Neglected personal projects (40 defects, 18.6%), excessive social media (34, 15.8%), and not applying for jobs (33, 15.3%) account for half of all defects. Adding skipped exercise, short sleep, and not reading brings the total to 79.5%. Two of the top three are career habits, which is where improvement effort has the most leverage. At the other end, skipped meals (6) and lack of social interaction (10) are infrequent and are not priorities for the next cycle.

## 3.2 Habit-level trends

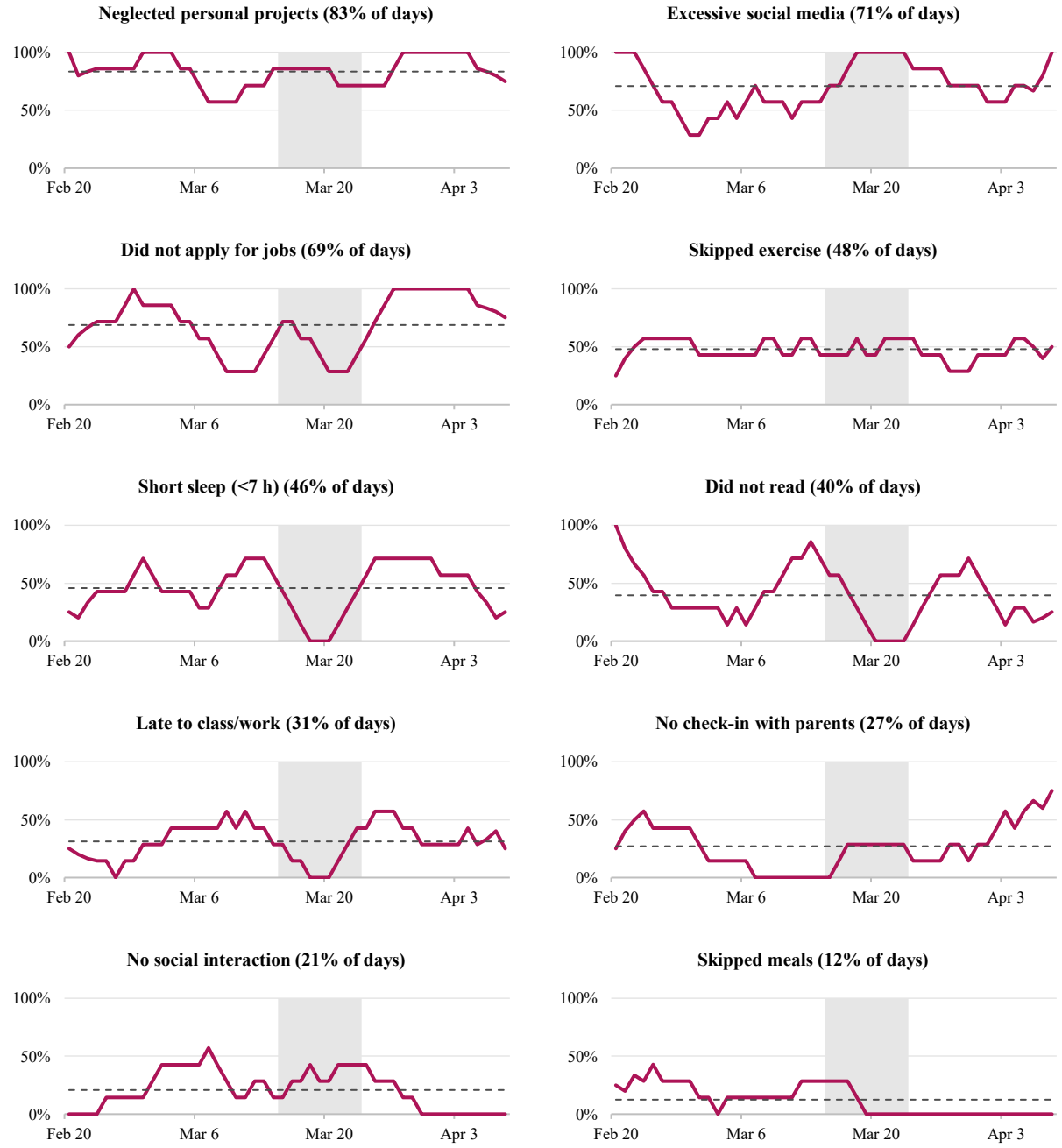



**Figure 3.** Seven-day centered rolling defect rate for each habit (solid line). The dashed line is the habit's overall rate, and the gray band marks Spring Break.

- **Excessive social media** rose from 57% of days before the break to 100% during it, and stayed elevated afterward (75%).
- **Short sleep and lateness** dropped to zero during the break and returned as soon as classes resumed (63% and 44% of post-break days).
- **Job applications** were missed on 94% of post-break days, up from 61% before the break, while **personal projects** were the most persistent defect in every period (78% to 88%).

- **Reading** improved during the break (missed on 22% of days versus 48% before), as the extra free time made room for it.
- **Skipped meals** stopped entirely after March 15 (0 of the final 24 days), reflecting a more consistent meal routine in the second half of the study.
- **Isolation and missed check-ins with parents** tended to coincide with exam periods, when long study sessions crowded out social time. Exam dates were not logged, so this is treated as a hypothesis.

### 3.3 Stratification: day of week and academic calendar

Stratifying the data by time factors separates two different kinds of defects: those driven by workload and those driven by unstructured time.

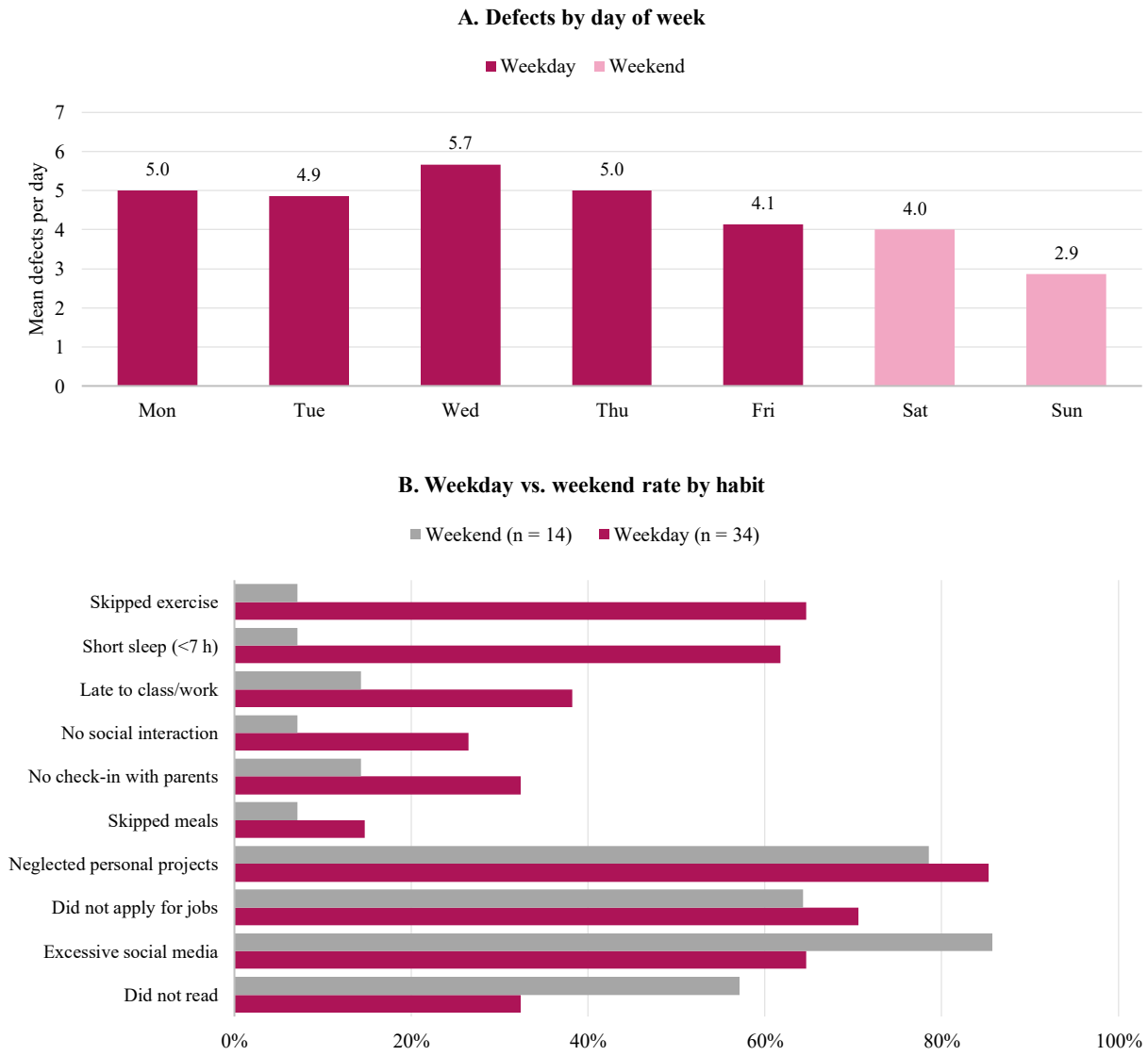



**Figure 4.** Stratification by day of week. (A) Mean defects per day, with weekends in light pink. (B) Share of days with each defect on weekdays versus weekends, sorted by the size of the gap.

- **Weekdays are the main source of defects.** Weekdays average 4.9 defects and weekends 3.4 (Mann-Whitney U test, $p = 0.004$). Wednesday is the worst day (5.7) and Sunday the best (2.9).

- **Workload defects are weekday defects.** Skipped exercise (65% of weekdays vs. 7% of weekends), short sleep (62% vs. 7%), and lateness (38% vs. 14%) almost vanish on weekends.
- **Exercise follows the class timetable.** My dance classes fall between Thursday and Sunday; exercise was skipped on 85% of Monday to Wednesday days and only 21% of Thursday to Sunday days.
- **Free-time defects rise on weekends.** Excessive social media (86% vs. 65%) and not reading (57% vs. 32%) are more common on weekends, so weekends trade workload defects for free-time defects.

**Table 4.** Defect rates by academic period (share of days with each defect)

| Habit | Pre-break (Feb 20 to Mar 14, n = 23) | Spring Break (Mar 15 to 23, n = 9) | Post-break (Mar 24 to Apr 8, n = 16) |
|---|---|---|---|
| **Mean defects per day** | **4.48** | **3.56** | **5.00** |
| Neglected personal projects | 83% | 78% | 88% |
| Excessive social media | 57% | 100% | 75% |
| Did not apply for jobs | 61% | 44% | 94% |
| Skipped exercise | 48% | 44% | 50% |
| Short sleep (<7 h) | 52% | 0% | 63% |
| Did not read | 48% | 22% | 38% |
| Late to class/work | 35% | 0% | 44% |
| No check-in with parents | 22% | 22% | 38% |
| No social interaction | 22% | 33% | 13% |
| Skipped meals | 22% | 11% | 0% |

Average defects fell from 4.48 per day before the break to 3.56 during it, then rose to 5.00 afterward. The difference is directional but not statistically significant at the 5% level (Kruskal-Wallis $H = 4.62$, $p = 0.10$), partly because the break contains only nine days, four of which fall on weekends. The more important pattern is the change in the defect mix: the break removed schedule-driven defects and replaced them with free-time defects.

## 3.4 Correlation analysis

Because every habit is binary, relationships were measured with the phi coefficient (Pearson's r applied to 0/1 data). With $n = 48$ days, $|r| \geq 0.285$ is significant at $\alpha = 0.05$. With 45 habit pairs, about two pairs would pass that bar by chance, so the interpretation focuses on the strongest relationships ($p < 0.01$).




**Figure 5.** Correlation matrix of the ten habit defects (phi coefficients). Pink cells indicate defects that tend to occur on the same day; gray cells indicate defects that trade off. Bold values are significant at $p < 0.05$.

**Table 5.** Strongest relationships, with conditional rates for interpretation

| Habit pair | r | p | What it means in practice |
|---|---|---|---|
| Short sleep & skipped exercise | 0.46 | 0.001 | Exercise skipped on 73% of short-sleep days vs. 27% of other days |
| No social interaction & skipped exercise | 0.43 | 0.002 | Exercise skipped on 90% of isolated days vs. 37% of other days |
| Excessive social media & short sleep | −0.42 | 0.003 | Short sleep on 32% of high-social-media days vs. 79% of other days |
| No check-in with parents & did not read | −0.40 | 0.005 | Reading missed on 8% of no-call days vs. 51% of other days |
| Short sleep & late to class/work | 0.37 | 0.009 | Late on 50% of short-sleep days vs. 15% of other days |
| Did not apply for jobs & neglected projects | 0.30 | 0.037 | Projects neglected on 91% of no-application days vs. 67% of other days |

The relationships form four interpretable patterns:

- **A workload cluster with sleep at the center.** Short sleep, skipped exercise, lateness, and isolation occur together on demanding weekdays. Sleep is connected to the most other defects, which makes it the highest-leverage habit to improve.
- **Social media reflects free time rather than lost sleep.** The negative correlation means high-scroll days were *well-rested* days: both happen on low-pressure days such as weekends and the break. A lagged check (social media on one day versus short sleep the next) found no relationship either ($r = 0.02$, $p = 0.87$). This contradicted my starting assumption and shifts the improvement focus from sleep hygiene to how unstructured time is used.

- **Evening activities compete.** Calling my parents and reading rarely fail on the same day; on days without a call, reading was missed only 8% of the time. The two habits appear to compete for the same evening window.
- **Career habits fail together.** Job applications and personal projects are both deferred under the same conditions, consistent with their nearly identical trend lines.

*Correlation describes same-day co-occurrence and does not establish causation; these findings are inputs to the root-cause analysis below rather than conclusions on their own.*

### 3.5 Root-cause analysis: fishbone (Ishikawa) diagram

The fishbone diagram brings the findings together. It organizes potential causes of the high defect rate into six categories adapted from the classic 6M framework: People, Methods, Technology (Machines), Schedule (Materials), Measurement, and Environment (Mother Nature). Filled markers are causes supported by evidence in Sections 3.2 to 3.4; hollow markers are hypotheses this dataset cannot test and should be checked in the next cycle.

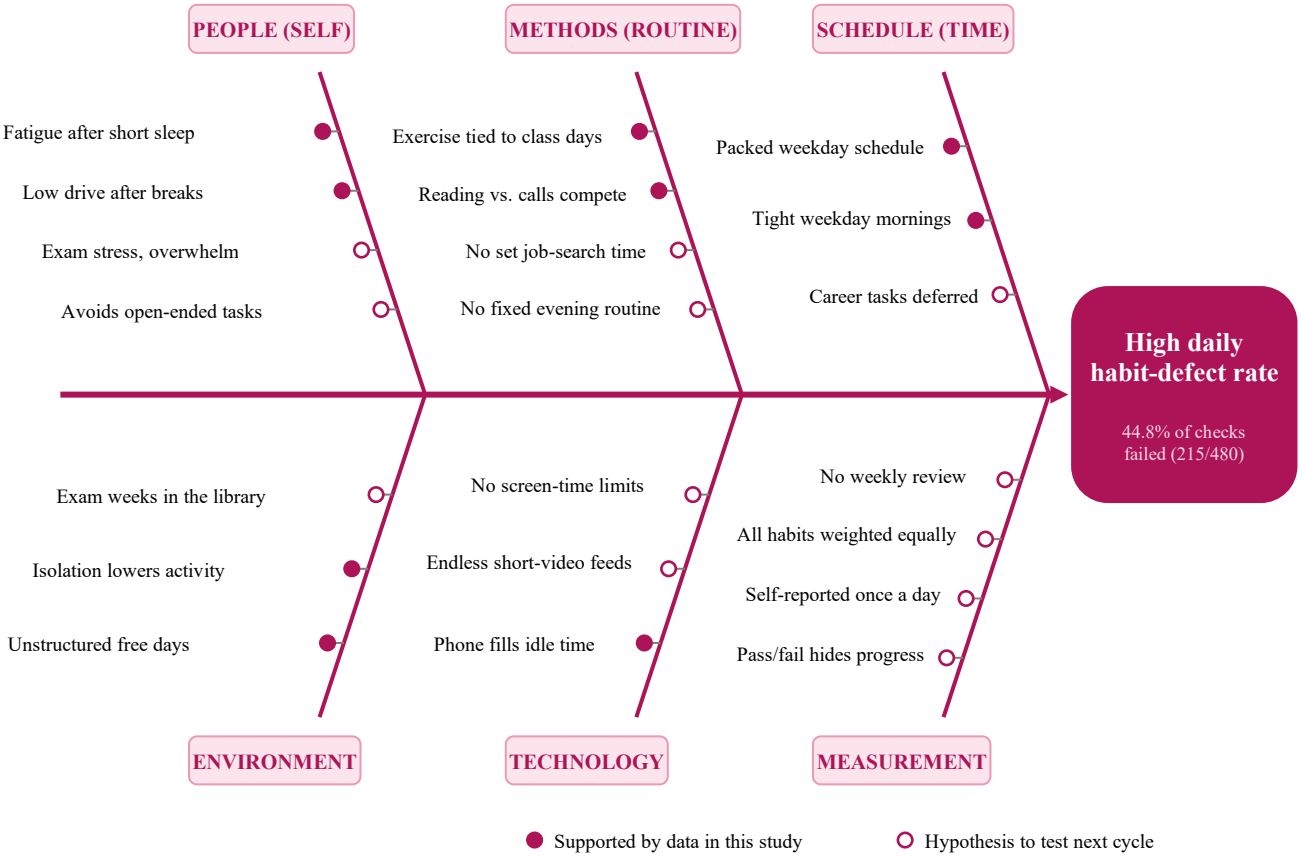



**Figure 6.** Cause-and-effect (Ishikawa) diagram for the daily habit-defect rate.

**Reading the diagram.** The evidence points to three root causes that sit upstream of most defects:

1. **Weekday schedule load (Schedule, Environment).** Classes and work fill weekdays, which drives short sleep and, through it, lateness and skipped exercise.
2. **No protected time for career work (Methods, People).** Job applications and personal projects have no fixed slot, so they are deferred whenever the day gets busy, which on weekdays is most of the time.

3. **Unstructured time defaults to the phone (Technology, Environment).** On weekends and breaks the workload defects disappear, but idle time fills with social media instead of planned activities.

The Measurement branch is a reminder that the measurement system itself shapes the results. A binary threshold records four job applications the same as zero, and every habit counts equally even though their consequences differ. These limitations are addressed in Section 8.

# 4. Statistical process control

## 4.1 Chart selection and assumptions

Each day is a rational subgroup of n = 10 binary checks, so the data are attribute data and the **P chart** (with its count equivalent, the NP chart) is the primary tool. $\bar{X}$ and S and individuals (I-MR) charts were included as robustness checks, and **CUSUM** and **EWMA** charts were added because they detect small, sustained shifts that Shewhart charts can miss.

The P chart assumes daily defect counts follow a binomial distribution. The observed variance of daily counts (2.51) is almost identical to the binomial expectation of $10\bar{p}(1-\bar{p}) = 2.47$, a dispersion ratio of 1.02, so there is no evidence of over-dispersion and binomial limits are appropriate.

## 4.2 P and NP charts

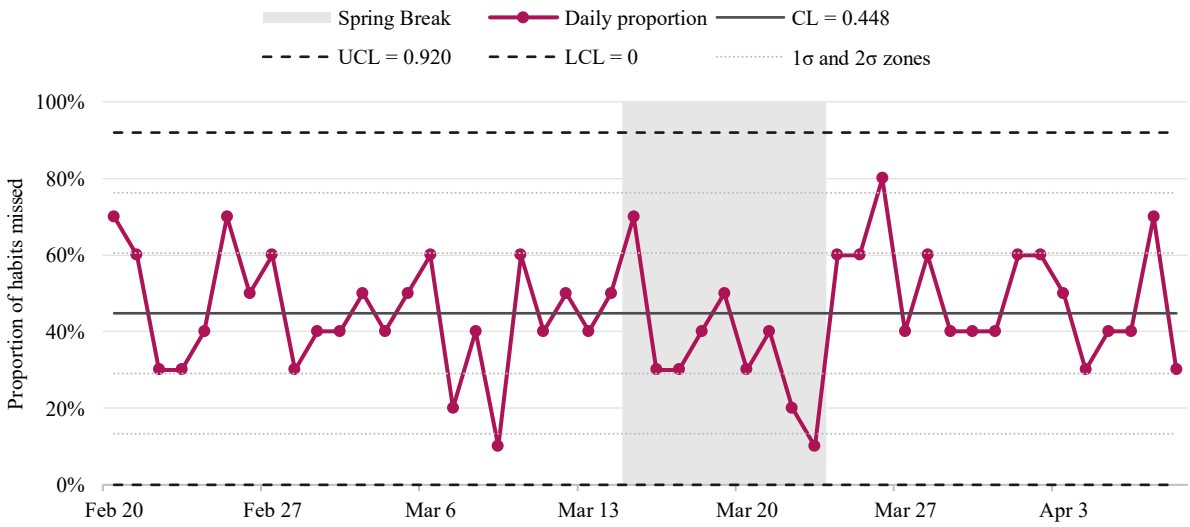



**Figure 7.** P chart of the daily defect proportion (n = 10). Dotted gray lines mark the 1σ and 2σ zones used for the Western Electric run rules; the gray band marks Spring Break.

With $\bar{p} = 0.448$ and $\sigma p = 0.157$, the limits are UCL = 0.920 and LCL = 0 (the calculated value of −0.024 is truncated at zero). All 48 points fall inside the limits, and none of the Western Electric run rules are triggered (2 of 3 points beyond 2σ, 4 of 5 beyond 1σ, 8 consecutive points on one side, or 6 points trending). The most extreme points are March 26 (z = +2.24) and March 9 and 23 (z = −2.21). Because the subgroup size is constant, the NP chart is the same chart rescaled by 10 (CL = 4.48, UCL = 9.20, LCL = 0) and gives identical conclusions.

## 4.3 X̄ and S charts

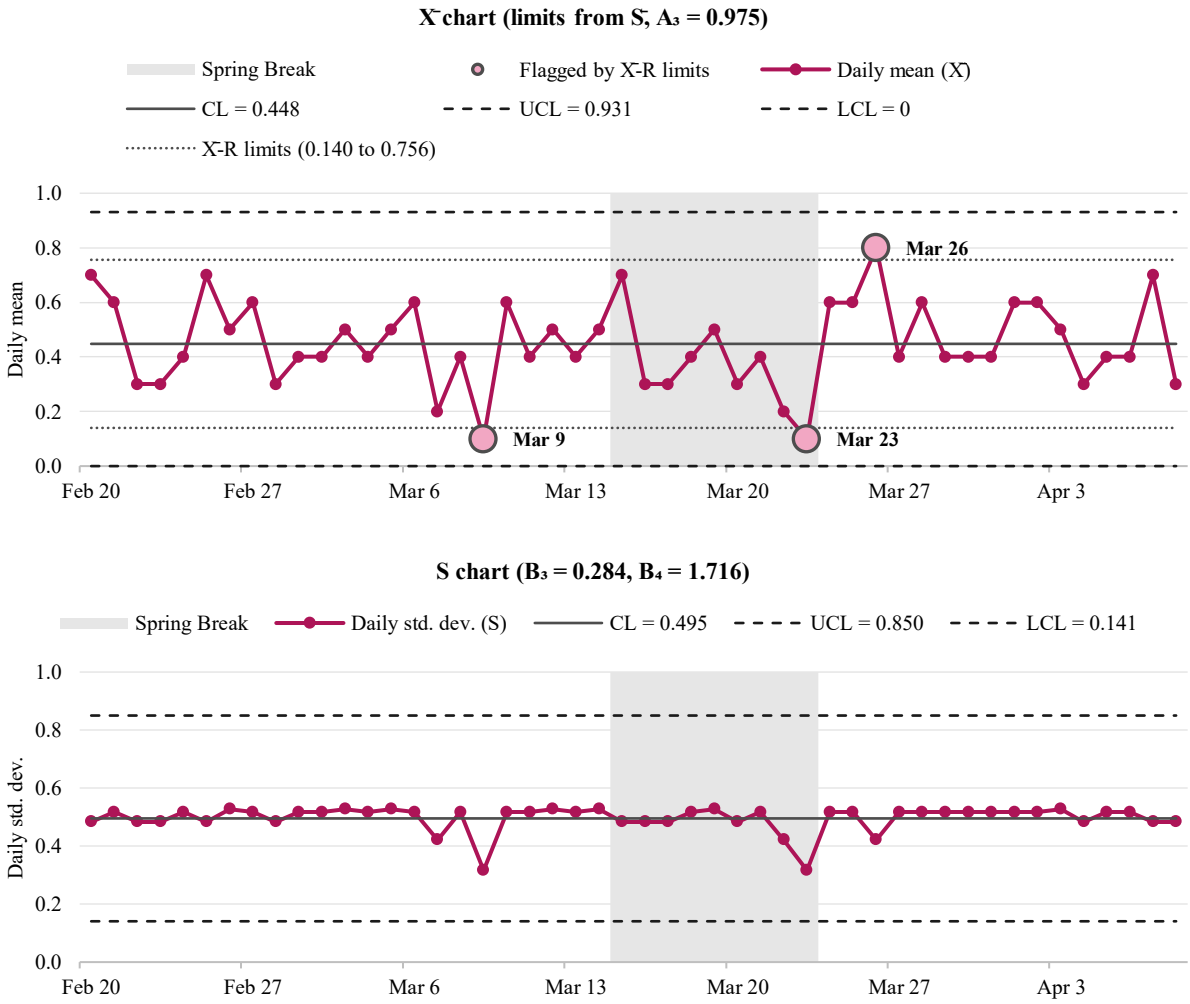



**Figure 8.** X̄ chart (top) and S chart (bottom). The dotted lines on the X̄ chart show the narrower limits produced by the X̄-R method, with the three points they flag circled.

Using $\bar{S} = 0.495$ and the constants for n = 10, the X̄ chart limits are 0 to 0.931 and the S chart limits are 0.141 to 0.850. Every point is within both sets of limits.

**Methodology note: why the X̄-R chart produced false alarms**

An initial X̄-R analysis flagged three days as out of control: March 9 and March 23 (0.10) and March 26 (0.80). Investigating this showed the signals were an artifact of applying range-based limits to 0/1 data.

For binary data, any day with at least one pass and one fail has a range of exactly 1, so $\bar{R} = 1.00$ regardless of how variable the process really is. The resulting half-width, $A_2\bar{R} = 0.308$, understates the true 3σ width of a daily mean (0.472 under the binomial model). The S-based limits ($A_3\bar{S} = 0.483$) and the P chart agree closely with each other, and both show the process in control.

The S chart has a related property: for 0/1 data, a day's standard deviation is determined entirely by its proportion, $s = \sqrt{(10p(1-p)/9)}$. Its lowest points (0.316 on March 9 and 23) are simply the days with p = 0.10, so the S chart adds no information beyond the P chart. The practical lesson: **match the chart to the data type**, and treat variables charts on attribute data as a check rather than the primary analysis.

## 4.4 Individuals and moving range (I-MR) charts

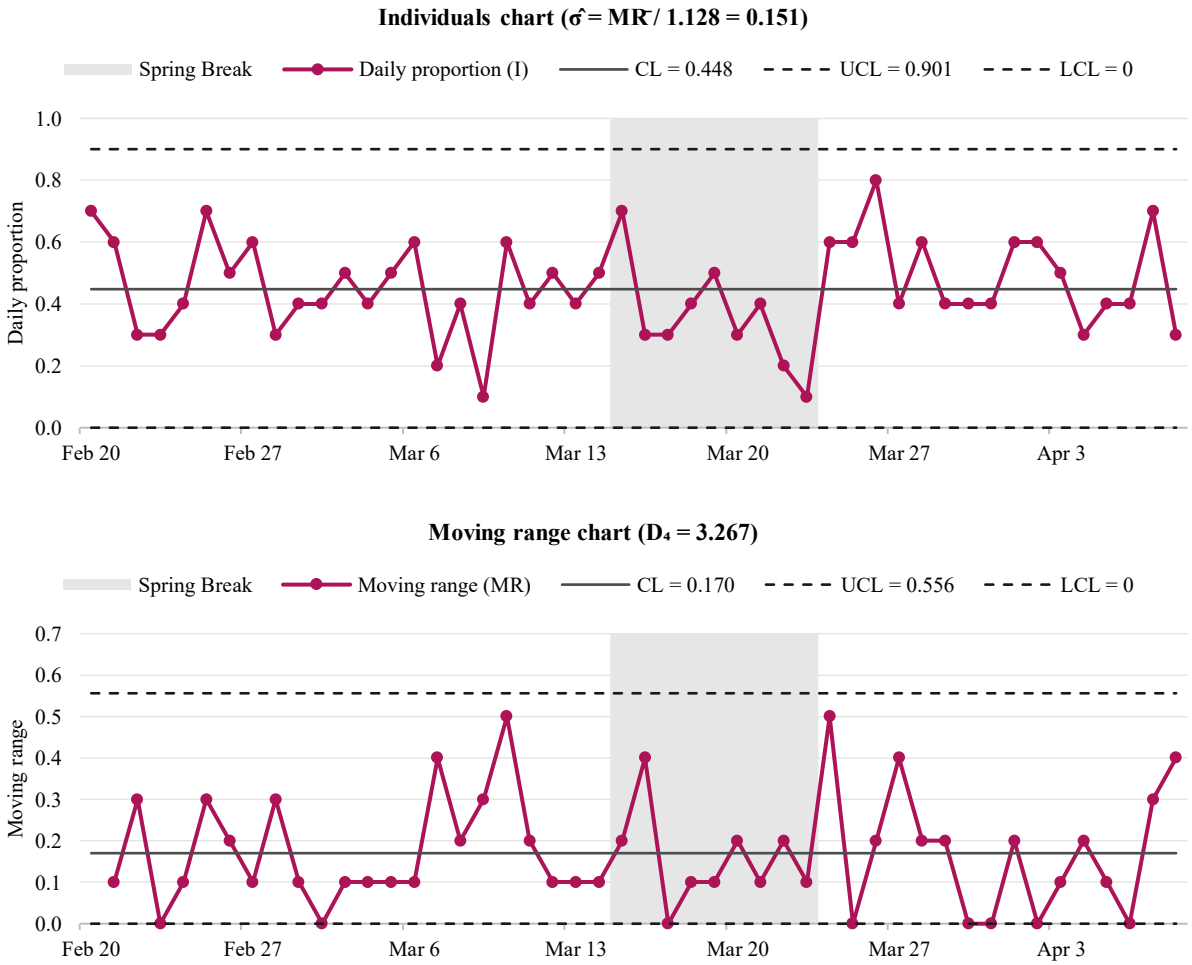



**Figure 9.** Individuals chart of the daily proportion (top) and moving range chart (bottom).

The I-MR chart estimates variation from consecutive days rather than from the binomial model. Its estimate, $\hat{\sigma} = \bar{MR} / 1.128 = 0.151$, is very close to the binomial $\sigma p$ of 0.157, which independently confirms the P chart limits. The individuals limits are 0 to 0.901 and the moving range UCL is 0.556. The largest moving ranges (0.50) occur on March 10 and March 24, both Monday rebounds from a Sunday low, which is the weekday effect from Section 3.3 showing up in the day-to-day swings. No point exceeds its limit, and just over half of the moving ranges (25 of 47) sit below the center line of 0.170.

## 4.5 CUSUM chart

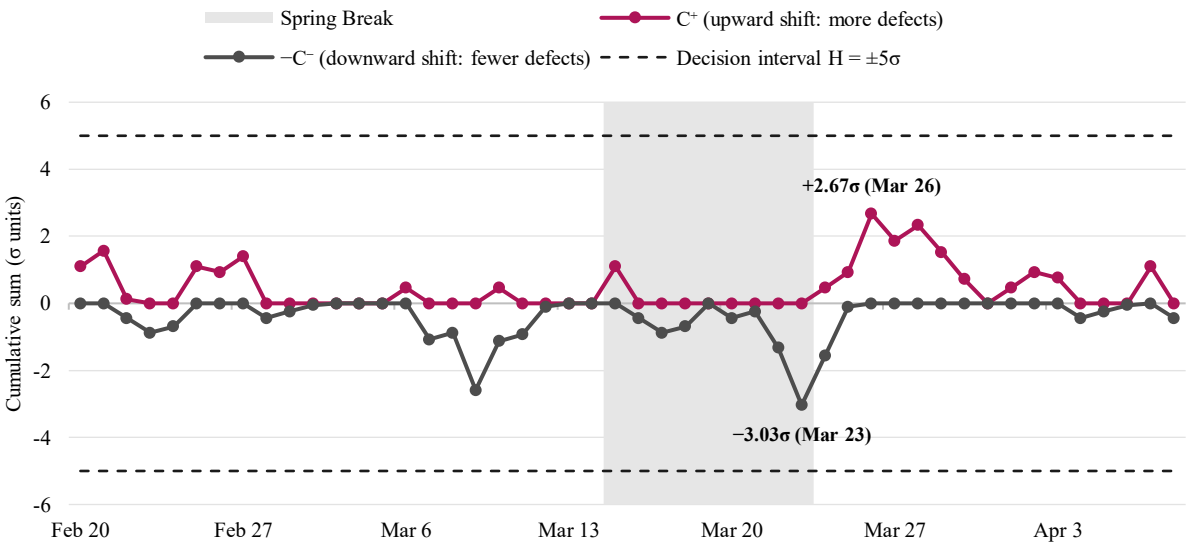



**Figure 10.** Tabular CUSUM in standardized units (k = 0.5, h = 5). The upper statistic accumulates evidence of more defects; the lower statistic (plotted below zero) accumulates evidence of fewer.

The tabular CUSUM was standardized using σp = 0.157, with reference value k = 0.5σ (tuned to detect a 1σ shift, about 1.6 defects per day) and decision interval h = 5σ, which gives an in-control average run length of roughly 465 days. Standardizing matters: applied to the raw 0 to 1 proportion scale, a threshold of h = 5 would correspond to about 32σ and could never signal.

No signals occurred. The largest excursions line up with the academic calendar: the lower CUSUM built to −3.03σ on March 23 as below-average days accumulated through Spring Break, and the upper CUSUM rose to +2.67σ on March 26 after classes resumed. Both reset to zero within a week, so the break caused a temporary dip rather than a sustained shift in the process.

## 4.6 EWMA chart

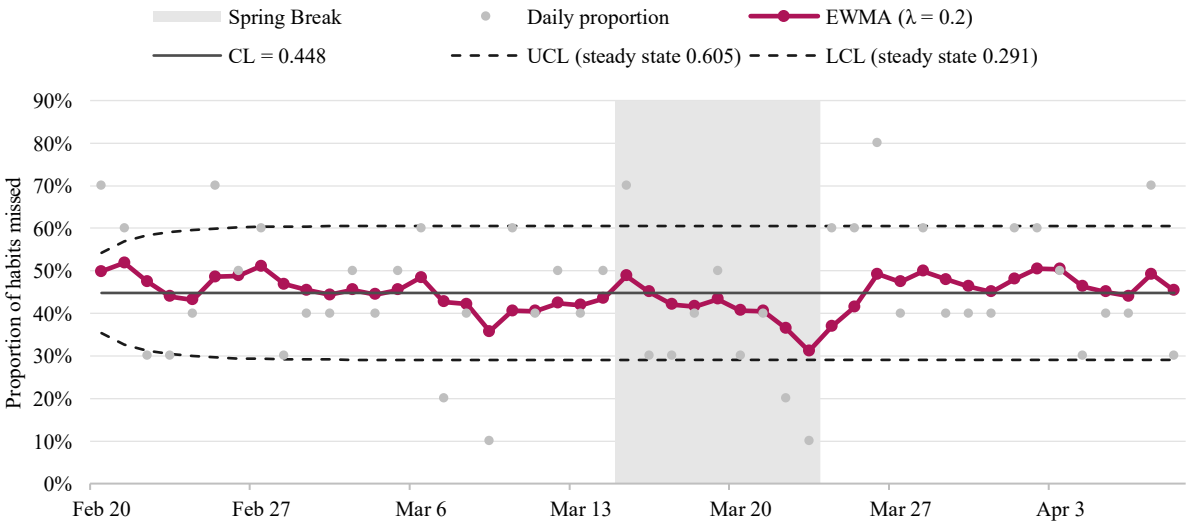



**Figure 11.** EWMA chart (λ = 0.2, L = 3) with time-varying control limits. Gray dots show the raw daily proportions.

The EWMA weights recent days more heavily (λ = 0.2), making it sensitive to small, gradual drifts. Its limits widen from ±0.094 on the first day to a steady state of 0.291 to 0.605. The EWMA ranged from 0.311 (March 23) to 0.519 (February 21) and never crossed a limit. Its closest approach to the lower limit was again the end of Spring Break, consistent with the CUSUM.

## 4.7 Moving averages

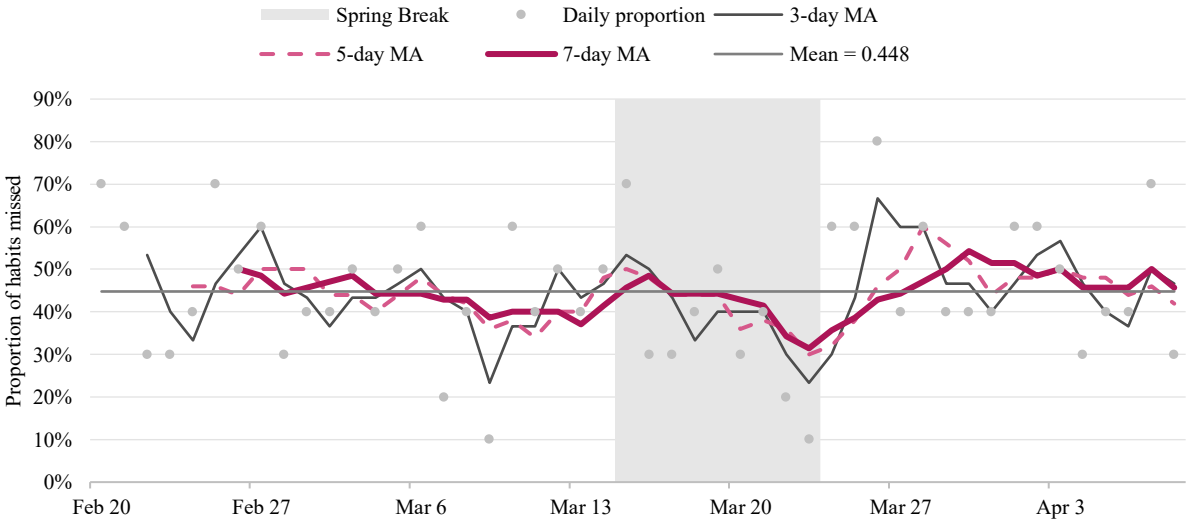



**Figure 12.** Three-, five-, and seven-day moving averages of the daily defect proportion.

The 3-day average reacts to individual days, swinging from 0.233 (March 9) to 0.667 (March 26). The 7-day average stays between 0.314 and 0.543. Because every 7-day window contains exactly one weekend, it removes the weekday effect entirely, which makes it the best smoothing window for routine monitoring. The rapid recovery of all three averages after late March shows that bad stretches self-correct within days rather than persisting.

## 4.8 Summary of control chart results

**Table 6.** Control chart summary

| Chart | Parameters | Center | Limits | Signals | Verdict |
|---|---|---|---|---|---|
| P | n = 10 | 0.448 | 0 to 0.920 | None (incl. run rules) | In control |
| NP | n = 10 | 4.48 | 0 to 9.20 | None | In control |
| $\bar{X}$ (S-based) | $A_3 = 0.975$ | 0.448 | 0 to 0.931 | None | In control |
| S | $B_3 = 0.284$, $B_4 = 1.716$ | 0.495 | 0.141 to 0.850 | None | In control |
| Individuals | $\hat{\sigma} = 0.151$ | 0.448 | 0 to 0.901 | None | In control |
| Moving range | $D_4 = 3.267$ | 0.170 | 0 to 0.556 | None | In control |
| CUSUM | k = 0.5σ, h = 5σ | 0 | ±5σ | None (peak −3.03σ) | In control |
| EWMA | λ = 0.2, L = 3 | 0.448 | 0.291 to 0.605 | None | In control |
| $\bar{X}$ (R-based) | $A_2 = 0.308$ | 0.448 | 0.140 to 0.756 | 3 points | Method artifact |

**Conclusion of the SPC analysis.** The routine is a stable, predictable process with a high baseline defect rate. In quality terms, the variation is common-cause: individual bad days are not special events that need

separate explanations. Meaningful improvement therefore requires changing the system (schedule, routines, and environment) rather than reacting to single days, and the limits above provide the baseline against which future changes can be judged.

## 5. Improve (proposed countermeasures)

The countermeasures below map each priority defect to the root causes identified in Section 3. They are proposals for the next DMAIC cycle; targets are set relative to this baseline and will be validated with new data.

**Table 7.** Countermeasure plan

| Target defects | Root cause | Countermeasure | Success measure |
|---|---|---|---|
| Neglected personal projects; did not apply for jobs | No protected time for career work; tasks deferred under load | Two calendar-blocked 45-minute career sessions on weekday mornings; batch applications on lighter Thursday to Sunday schedules | Average rate across the two habits falls from 76% to below 50% of days |
| Short sleep, and through it lateness and skipped exercise | Weekday schedule load; tight mornings | Fixed weekday lights-out time; phone charges outside the bedroom; next-day preparation the night before | Weekday short-sleep rate falls from 62% to below 30% |
| Skipped exercise | Exercise depends on Thursday to Sunday classes | One short Monday to Wednesday workout (for example, a 20-minute session or a swim) | Monday to Wednesday skip rate falls from 85% to below 50% |
| Excessive social media | Unstructured time defaults to the phone | Daily app limit of 60 minutes; plan one weekend activity in advance; swap scrolling time for reading | Weekend social media rate falls from 86% to below 50% |
| Did not read; no check-in with parents | Both compete for the same evening window | Move parent calls to commute or walking time; keep a fixed reading slot before bed | Both habits completed on the same day at least 4 days a week (baseline 2.5) |
| Defect spikes around breaks | Loss of structure, then a hard re-entry | Keep two anchor habits (sleep and one career block) during breaks; plan the first week back in advance | No CUSUM excursion beyond ±3σ around the next break |

# 6. Control (proposed control plan)

Sustaining any improvement requires a monitoring routine with clear reaction rules. Because the tracking sheet is already in place, the control plan adds a weekly review rather than new data collection.

**Table 8.** Control plan

| What is monitored | Method | Frequency | Reaction rule |
|---|---|---|---|
| Daily defect proportion | P chart using the baseline limits (CL 0.448, UCL 0.920) | Updated daily, reviewed every Sunday | A point beyond the limits or a run-rule violation triggers a written note on the likely special cause |
| Small sustained shifts | EWMA ($\lambda = 0.2$) and 7-day moving average | Weekly | An EWMA point below the LCL after a change counts as confirmed improvement; recalculate limits from the new data |
| Defect mix | Weekly Pareto of the ten habits | Weekly | A habit missed on more than 70% of days for two consecutive weeks triggers a review of its countermeasure |
| High-risk periods | Calendar flags for exams and breaks | Before each period | Activate the break and exam routine: anchor habits only, with reduced but non-zero targets |

# 7. Conclusions

This project treated a personal routine as a measurable process and found that it behaves like one. Defects are concentrated in a small number of habits, they are driven by identifiable structural factors (above all the weekday schedule), and the overall process is statistically stable. That stability is useful: it means the routine is predictable, and it sets a clear baseline against which future changes can be tested.

Three lessons stand out. First, **data can contradict intuition.** I expected social media to be eroding my sleep, but the data show the opposite pattern, which redirects effort toward how free time is structured. Second, **method choice matters.** The same data looked out of control under an $\bar{X}$-R chart and in control under the charts suited to binary data; understanding why is as important as producing the chart. Third, **stable is not the same as good.** A 44.8% defect rate is consistent, but improving it requires deliberate changes to the system, not more effort on individual days.

# 8. Limitations and next steps

- **Single subject and short window.** Forty-eight days from one person limits generalization and the power of subgroup comparisons such as the Spring Break test.
- **Binary thresholds lose information.** Recording hours slept, minutes on social media, and the number of applications would allow true variables charts and more sensitive analysis.
- **Equal weighting.** A weighted demerit system (for example, a U chart of demerits per day) would reflect that some habits matter more than others.
- **Unmeasured drivers.** Exam dates, workload, mood, and stress were not recorded; adding them as stratification variables would test the hypotheses in the fishbone diagram.
- **Self-reporting and multiple comparisons.** Records are self-reported once a day, and the correlation analysis tests 45 pairs, so weaker relationships should be confirmed with new data.

- **Phase II monitoring.** The next step is to implement the countermeasures in Section 5, collect a new period of data, and compare it with this baseline using the control plan and a two-proportion test.

## Appendix A. Daily check sheet data

1 = defect recorded. Column numbers match the habit numbers in Table 2.

| Date | Day | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Feb 20 | Thu | | 1 | 1 | 1 | | | 1 | 1 | 1 | 1 | 7 |
| Feb 21 | Fri | 1 | 1 | | | | 1 | | 1 | 1 | 1 | 6 |
| Feb 22 | Sat | | 1 | | | | | | | 1 | 1 | 3 |
| Feb 23 | Sun | | 1 | | | | | | | 1 | 1 | 3 |
| Feb 24 | Mon | | 1 | 1 | | | | 1 | 1 | | | 4 |
| Feb 25 | Tue | | 1 | 1 | 1 | | 1 | 1 | 1 | 1 | | 7 |
| Feb 26 | Wed | | | 1 | 1 | | | 1 | 1 | 1 | | 5 |
| Feb 27 | Thu | | | 1 | 1 | 1 | 1 | | 1 | 1 | | 6 |
| Feb 28 | Fri | | | | | | | | 1 | 1 | 1 | 3 |
| Mar 1 | Sat | 1 | 1 | | | | | | 1 | 1 | | 4 |
| Mar 2 | Sun | | | | 1 | | | | 1 | 1 | 1 | 4 |
| Mar 3 | Mon | 1 | | 1 | 1 | | | 1 | | 1 | | 5 |
| Mar 4 | Tue | | 1 | | | 1 | | | 1 | 1 | | 4 |
| Mar 5 | Wed | | 1 | 1 | | 1 | | | 1 | 1 | | 5 |
| Mar 6 | Thu | 1 | | 1 | 1 | 1 | | | 1 | 1 | | 6 |
| Mar 7 | Fri | | 1 | | | | 1 | | | | | 2 |
| Mar 8 | Sat | 1 | | | | | | | 1 | 1 | 1 | 4 |
| Mar 9 | Sun | | 1 | | | | | | | | | 1 |
| Mar 10 | Mon | 1 | 1 | 1 | 1 | 1 | | | | | 1 | 6 |
| Mar 11 | Tue | | | 1 | 1 | | | | | 1 | 1 | 4 |
| Mar 12 | Wed | 1 | 1 | 1 | 1 | | | | | 1 | | 5 |
| Mar 13 | Thu | | | | 1 | | | | 1 | 1 | 1 | 4 |
| Mar 14 | Fri | 1 | | | 1 | | 1 | | | 1 | 1 | 5 |
| Mar 15 | Sat | | 1 | 1 | | 1 | 1 | | 1 | 1 | 1 | 7 |
| Mar 16 | Sun | | 1 | | | | | | 1 | | 1 | 3 |
| Mar 17 | Mon | | 1 | | | | | | 1 | 1 | | 3 |
| Mar 18 | Tue | | 1 | 1 | | | | | 1 | 1 | | 4 |
| Mar 19 | Wed | | 1 | 1 | | 1 | | 1 | | 1 | | 5 |
| Mar 20 | Thu | | 1 | | | | | 1 | | 1 | | 3 |
| Mar 21 | Fri | | 1 | 1 | | 1 | | | | 1 | | 4 |
| Mar 22 | Sat | | 1 | | | | | | | 1 | | 2 |
| Mar 23 | Sun | | 1 | | | | | | | | | 1 |
| Mar 24 | Mon | 1 | 1 | 1 | 1 | 1 | | | 1 | | | 6 |
| Mar 25 | Tue | 1 | 1 | 1 | 1 | | | | 1 | 1 | | 6 |
| Mar 26 | Wed | 1 | 1 | 1 | 1 | 1 | | 1 | 1 | 1 | | 8 |
| Mar 27 | Thu | | | | 1 | | | | 1 | 1 | 1 | 4 |

| Date | Day | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Mar 28 | Fri | 1 | 1 | | 1 | | | | 1 | 1 | 1 | 6 |
| Mar 29 | Sat | | 1 | | | | | | 1 | 1 | 1 | 4 |
| Mar 30 | Sun | | 1 | | | | | | 1 | 1 | 1 | 4 |
| Mar 31 | Mon | | | | 1 | | | 1 | 1 | 1 | | 4 |
| Apr 1 | Tue | 1 | 1 | 1 | 1 | | | | 1 | 1 | | 6 |
| Apr 2 | Wed | | 1 | 1 | 1 | | | | 1 | 1 | 1 | 6 |
| Apr 3 | Thu | | | 1 | 1 | | | 1 | 1 | 1 | | 5 |
| Apr 4 | Fri | 1 | | | | | | | 1 | 1 | | 3 |
| Apr 5 | Sat | | 1 | | | | | 1 | 1 | 1 | | 4 |
| Apr 6 | Sun | | 1 | | | | | 1 | 1 | 1 | | 4 |
| Apr 7 | Mon | 1 | 1 | 1 | 1 | | | | 1 | 1 | 1 | 7 |
| Apr 8 | Tue | | 1 | 1 | | | | 1 | | | | 3 |
| **Total** | | **15** | **34** | **23** | **22** | **10** | **6** | **13** | **33** | **40** | **19** | **215** |

# Appendix B. Formulas and constants

**Table 9.** Control chart formulas used in this report (n = 10 checks per day, m = 48 days)

| Chart | Center line and limits |
|---|---|
| P | $\bar{p} = \Sigma\text{defects} \div (n \cdot m)$; limits = $\bar{p} \pm 3\sqrt{(\bar{p}(1-\bar{p})/n)}$ |
| NP | $n\bar{p}$; limits = $n\bar{p} \pm 3\sqrt{(n\bar{p}(1-\bar{p}))}$ |
| $\bar{X}$ and S | $\bar{X}$ limits = $\bar{p} \pm A_3\bar{S}$; S limits = $B_3\bar{S}$ and $B_4\bar{S}$; for n = 10: $A_3 = 0.975$, $B_3 = 0.284$, $B_4 = 1.716$ |
| $\bar{X}$ and R | $\bar{X}$ limits = $\bar{p} \pm A_2\bar{R}$; for n = 10: $A_2 = 0.308$ (shown for comparison only) |
| I-MR | $\hat{\sigma} = M\bar{R} / d_2$ with $d_2 = 1.128$; I limits = $\bar{p} \pm 3\hat{\sigma}$; MR UCL = $D_4M\bar{R}$ with $D_4 = 3.267$ |
| CUSUM | $z(i) = (p(i) - \bar{p})/\sigma p$; $C^+(i) = \max[0, z(i) - k + C^+(i-1)]$; $C^-(i) = \max[0, -z(i) - k + C^-(i-1)]$; signal if either exceeds h; k = 0.5, h = 5 |
| EWMA | $z(i) = \lambda p(i) + (1-\lambda)z(i-1)$, with $z(0) = \bar{p}$; limits = $\bar{p} \pm L\sigma p\sqrt{(\lambda/(2-\lambda) \times [1 - (1-\lambda)\text{^}(2i)])}$; $\lambda = 0.2$, L = 3 |
| Phi coefficient | Pearson r on 0/1 data; $t = r\sqrt{((m-2)/(1-r^2))}$ with $m - 2 = 46$ degrees of freedom |
| Process sigma | DPMO = $10^6 \times$ defects ÷ opportunities; sigma level = $\Phi^{-1}(1 - \text{DPO}) + 1.5$ |